

INFLUENZA

A single mutation in bovine influenza H5N1 hemagglutinin switches specificity to human receptors

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In 2024, several human infections with highly pathogenic clade 2.3.4.4b bovine influenza H5N1 viruses in the United States raised concerns about their capability for bovine-to-human or even human-to-human transmission. In this study, analysis of the hemagglutinin (HA) from the first-reported human-infecting bovine H5N1 virus (A/Texas/37/2024, Texas) revealed avian-type receptor binding preference. Notably, a Gln²²⁶Leu substitution switched Texas HA binding specificity to human-type receptors, which was enhanced when combined with an Asn²²⁴Lys mutation. Crystal structures of the Texas HA with avian receptor analog LSTa and its Gln²²⁶Leu mutant with human receptor analog LSTc elucidated the structural basis for this preferential receptor recognition. These findings highlight the need for continuous surveillance of emerging mutations in avian and bovine clade 2.3.4.4b H5N1 viruses.

Since the first highly pathogenic avian influenza (HPAI) H5N1 virus A/goose/Guangdong/1/1996 emerged in 1996 (1), avian H5-subtype viruses have spread widely in Europe, Africa, North America, and Asia through migratory birds, resulting in diverse phylogenetic clades and subclades (2–8). In late 2021, an H5N1 clade 2.3.4.4b virus was detected in North America (9–12) and was able to infect a wide spectrum of avian species and mammals, including marine and terrestrial mammals and humans (13–25). In early 2024, the first human-infecting bovine H5N1 case involving close contact with infected dairy herds was reported in Texas, which heralded the outbreak of HPAI H5N1 virus in dairy cattle in the US. Genetic analysis of the influenza hemagglutinin (HA) indicated that this H5N1 virus (A/Texas/37/2024, Texas) belonged to H5 clade 2.3.4.4b (26). Since then, this virus has been detected in at least 282 dairy herds in 14 states in the US, with potential inter-species transmission (27, 28), including to farm cats, poultry, and humans.

As of the latest available data in October 2024, the US Centers for Disease Control and Prevention (CDC) has reported 15 human infections worldwide with the 2.3.4.4b virus (29) but 17 human infections with the H5 subtype in the US since 2022, including cases of the 2.3.4.4b virus from exposure to infected cows and poultry (30). Humans infected from cows and poultry have reported mild conjunctivitis and respiratory symptoms (26, 29, 31). Historically, however, sporadic human infections

from earlier clades of H5N1 from exposure to infected poultry resulted in 30% mortality rates in hospitalized cases (32), which is likely an overestimate, as it does not take into account infections that were minor or not reported. Although there are currently no documented cases of transmission between humans, there is concern that the virus could adapt for human-to-human transmission and potentially result in an influenza pandemic.

One barrier for transmission of avian viruses to humans is a mismatch in the receptor specificity of the HA that binds sialic acid receptors (33). The HA of avian influenza viruses recognizes “avian-type” receptors with sialic acid in an α 2-3 linkage (Neu5Ac α 2-3Gal). By contrast, the HA of human viruses recognizes “human-type” receptors with sialic acid in an α 2-6 linkage (Neu5Ac α 2-6Gal), which are abundantly expressed in the upper human airway (34). Acquisition of human-type receptor specificity is believed to be required for human-to-human transmission of influenza virus and is one of the major factors considered by the CDC for pandemic risk of a novel animal strain (33, 35, 36).

A switch in receptor binding preference of human viruses from avian or swine virus progenitors was previously shown for influenza pandemics in 1918, 1957, 1968, and 2009, when HA acquired Asp¹⁹⁰ and Asp²²⁵ (H3 numbering) in the H1 subtype and Leu²²⁶ and Ser²²⁸ in H2 and H3 subtypes (37–39). In the 1918 and 2009 H1N1 pandemics, there is some evidence for human-transmissible viruses with a single mutation (Glu¹⁹⁰Asp), which conferred human-type receptor specificity while retaining avian-type receptor specificity (dual specificity) (38, 40, 41). The sporadic human infections with the avian H5 subtype have therefore raised global health concerns as to whether H5 viruses could acquire the ability to switch

receptor preference. Several studies from the past decade have shown that Gln²²⁶Leu and Gly²²⁴Ser substitutions in avian H5 HA can promote binding to human receptors, although some binding to avian-type receptors is also retained (42–48). Aerosol transmission studies in ferrets have showed that additional mutations along with Leu²²⁶ or Ser²²⁸ can increase human receptor binding and concomitantly reduce avian-type receptor binding (49, 50). However, none of these mutants showed a complete switch to human receptors.

To assess the potential for the recent 2.3.4.4b viruses to acquire human-type receptor specificity, we introduced mutations into the receptor binding site (RBS) of the Texas HA protein and assessed receptor binding using surface plasmon resonance (SPR), enzyme-linked immunosorbent assay (ELISA), and glycan array analyses. Furthermore, we determined crystal structures for Texas H5 HA and its Gln²²⁶Leu mutant with avian and human receptor analogs to provide insights into the molecular basis of H5 HA receptor preference.

Results

Receptor specificity of Texas H5 HA and mutants

To examine glycan binding of the wild-type (WT) bovine Texas H5 HA, we used an SPR assay with extended linear avian-type α 2-3 (3SLN₃-L) and human-type α 2-6 sialosides (6SLN₃-L), which contain three N-acetylglucosamine (LacNAc, Gal β 1-4GlcNAc) repeats; an ELISA with linear and biantennary N-linked glycan sialosides with multiple LacNAc repeats (1 to 3); and a glycan array assay with a variety of α 2-3 and α 2-6 sialosides (Figs. 1 and 2 and figs. S1 to S4) (see supplementary materials and table S2). The SPR showed that WT Texas HA exhibits strong binding to α 2-3 SLN₃, with an apparent dissociation constant (K_D) of 138 nM, whereas no binding to α 2-6 SLN₃ was detected (K_D > 1 mM) (Fig. 1C). This result was consistent with the ELISA assay, which showed binding to linear α 2-3 sialosides (3SLN₁₋₃), as well as glycan array analysis, where binding was observed to a variety of linear and biantennary α 2-3 sialosides (Fig. 1, D and E). In all three assays, the WT Texas H5 HA exhibited strong avian-type specificity, as also reported recently by others (51, 52).

Notably, when we introduced a Gln²²⁶Leu substitution, binding specificity was completely switched from avian-type to human-type receptors in SPR, ELISA, and glycan array, with no detectable binding to α 2-3 sialosides (Fig. 2). In the ELISA, binding to biantennary N-linked glycan sialosides was stronger than to linear sialosides (Fig. 2B). This observation is also consistent with the glycan array analysis, which showed that the Gln²²⁶Leu mutant preferentially binds to α 2-6 sialylated biantennary glycans (Fig. 2C and table S2). Although the

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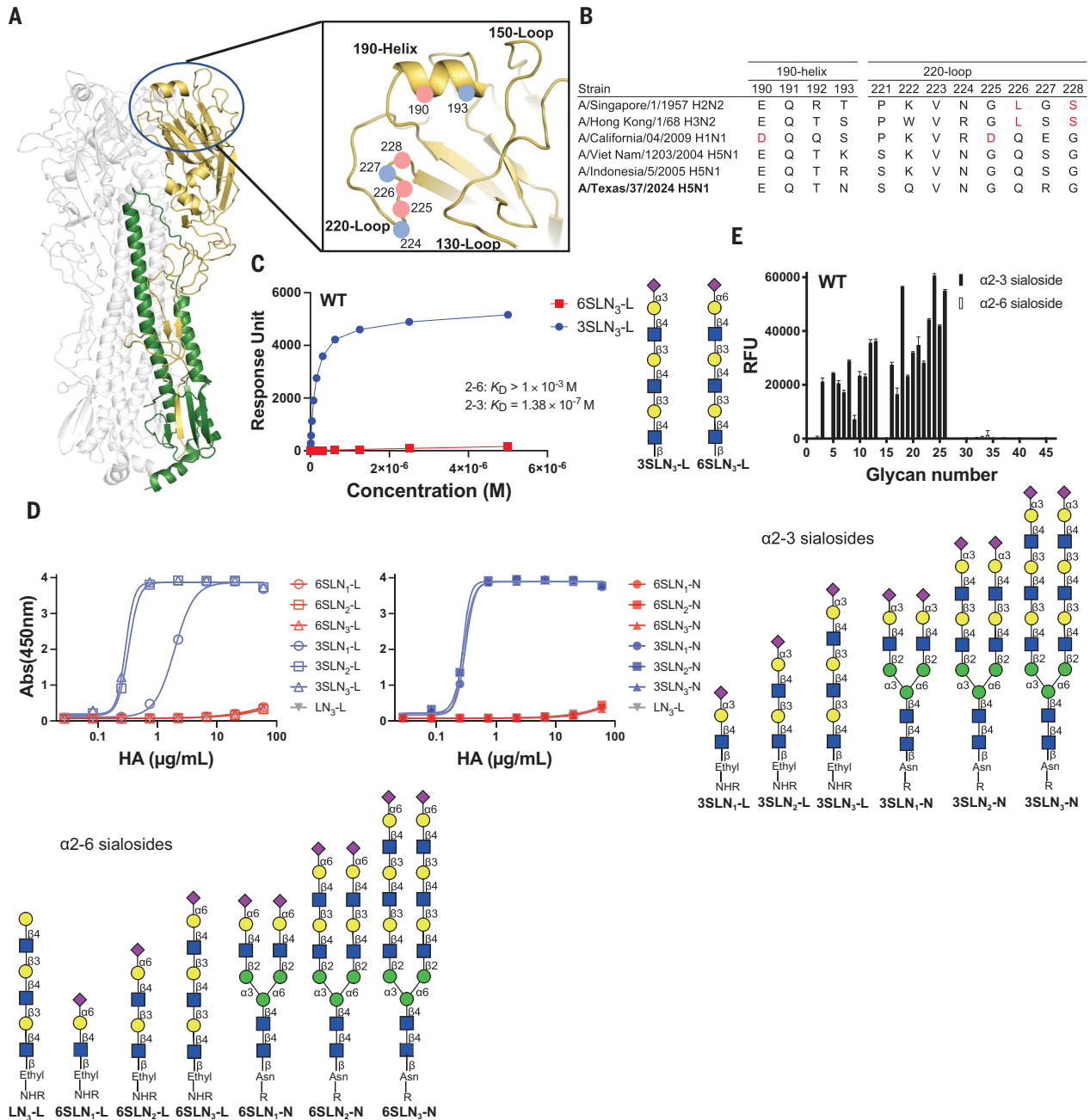


Fig. 1. Receptor characterization of WT bovine Texas H5 HA from A/Texas/37/2024 (H5N1). (A) The crystal structure of Apo Texas H5 HA was determined at 2.70-Å resolution (table S1). One HA monomer is highlighted in yellow for HA1 and in green for HA2. The receptor binding site is composed of the 130 loop, 150 loop, 190 helix, and 220 loop (see inset). Key residues and additional mutations analyzed in this study are shown as red and lavender dots, respectively. (B) Sequence comparison of 190 helix and 220 loop of WT Texas with human pandemic H1, H2, and H3 HA and human-infecting H5N1 HA. The hallmark residues for switch of receptor binding preference in H1, H2, and H3 are highlighted in red. (C) Binding affinity was measured by SPR. Response units are plotted against protein concentration. The binding curves representing

α 2-3 and α 2-6 are blue and red, respectively. (D) The α 2-3 and α 2-6 sialosides in the ELISA are based on linear (left, open shapes) and biantennary (right, solid shapes) N-linked glycans, each with one to three repeats of LacNAc (Gal β 1-4GlcNAc), as shown in the diagrams below. L, linear; N, N-linked; LN₃-L, a non-sialylated glycan; Abs, absorbance; R, -LCLC-Biotin. (E) Glycan microarray analysis of WT bovine Texas H5 HA. Two subsets of sialosides for α 2-3 (glycan 3 to 26, black bar) and α 2-6 (glycan 27 to 46, white bar) sialylated glycans imprinted on the chip are listed in table S2. RFU, relative fluorescence units. Single-letter abbreviations for the amino acid residues referenced throughout this paper are as follows: E, Glu; Q, Gln; R, Arg; T, Thr; P, Pro; S, Ser; D, Asp; G, Gly; N, Asn; K, Lys; W, Trp; V, Val; L, Leu; A, Ala; I, Ile.

binding of the Gln²²⁶Leu mutant HA to human-type receptors was weaker than the binding of the WT HA to α 2-3 sialosides, the change in specificity was nevertheless clear and pro-

nounced in all three assays. Moreover, in the ELISA, the Leu²²⁶ mutant bound more avidly to the α 2-6 sialylated N-linked glycans than did the HA from the human 2009 H1N1 swine

pandemic virus (A/California/04/2009, CA04) (Fig. 2B). To create the Leu²²⁶/Ser²²⁸ pairing that confers human-type receptor specificity in HAs of



Fig. 2. Receptor specificity of bovine Texas H5 HA with a Gln²²⁶Leu mutation and in combination with other mutations. The glycan receptor specificity for the H5 HA with Leu²²⁶, Leu²²⁶/Ser²²⁸, Lys²²⁴/Leu²²⁶, and Lys²²⁴/Leu²²⁶/Ser²²⁸ mutations in the RBS were measured by SPR (A), ELISA (B), and glycan array (C). A/California/04/09 (CA04) H1 HA was used as a positive control for binding to α -2-6 sialosides. (A) The K_D for each mutant in SPR assays was calculated based on the response units in figs. S1 and S2 using the Biacore

S200 evaluation software static affinity model. (B) Extended linear (left) and biantennary (right) α -2-3 and α -2-6 sialosides were used in the ELISA. The schematic structures of the glycans are represented in Fig. 1. (C) Texas HA Leu²²⁶ mutant binding to the glycan array. Glycan numbers 3 to 26 on the array are for α -2-3 sialosides (black bars), and 27 to 46 are for α -2-6 sialosides (white bars). The schematic shows main glycan structures represented in the binding to α -2-6 sialosides. Glycan numbers refer to the glycans shown in table S2.

H2N2 and H3N2 human viruses, we produced the Texas H5 HA with both the Gln²²⁶Leu and Gly²²⁸Ser mutations. This double mutant exhibited a specificity for α -2-6 sialosides, similar to the mutant with Gln²²⁶Leu alone (Fig. 2, A and B). However, substitution of Gly²²⁸Ser alone abrogated binding to both α -2-3 and α -2-6 sialosides in SPR and ELISA (figs. S1 to S3). We also produced HAs with Glu¹⁹⁰ to Asp¹⁹⁰ or Gly²²⁵ to Asp²²⁵, which are hallmark residues for the avian- to human-type switch in receptor binding in H1 pandemic viruses (33, 38). These mutations either eliminated binding to α -2-3 and α -2-6 sialosides (Asp¹⁹⁰) or retained avian-type binding preference (Asp²²⁵) (figs. S1 to S3), suggesting that they are not able to effect binding from avian- to human-type receptors in Texas H5 HA.

Some earlier investigations into the receptor specificity and transmission of H5N1 strains found that three HA mutations were required to effect a change to human-type receptor specificity (45, 49, 50). In one case, Asn²²⁴Lys was paired with Gln²²⁶Leu (49). When the Asn²²⁴Lys mutation was introduced into the Texas HA with either Leu²²⁶ or Leu²²⁶/Ser²²⁸, binding to extended linear α -2-6 sialosides by SPR and extended linear and biantennary N-linked α -2-6 sialosides by ELISA was enhanced, particularly to Lys²²⁴/Leu²²⁶ (Fig. 2, A and B). Weak binding to α -2-3 sialosides was found in the Lys²²⁴/Leu²²⁶/Ser²²⁸ triple mutant in comparison to the Leu²²⁶/Ser²²⁸ double mutant, which showed negligible binding to α -2-3 sialosides (Fig. 2, A and B). We also combined the Leu²²⁶ mutant with other mutations, including Asn¹⁹³Thr, Asn¹⁹³Lys, Arg²²⁷Asn, Arg²²⁷Ser, and Arg²²⁷Gln, which had previously shown improvement in H5 HA human-type receptor binding (53–55). However, none of these mutations exhibited notable increases in binding to α -2-6 sialosides when combined with the Leu²²⁶ mutant and, indeed, generally also eliminated binding to α -2-3 sialosides (figs. S1, S2, and S4). Binding of these mutants was predominantly to biantennary α -2-6 sialosides (fig. S4). Taken together, these results indicate that the Leu²²⁶ substitution from Gln²²⁶ is key to a switch in avian- to human-type receptor preference in Texas H5 HA.

Crystal structure of Texas H5 HA with avian receptor analog LSTa

To assess the structural basis of how mutations switch the receptor specificity in the Texas H5

HA, we first determined a crystal structure at 2.32-Å resolution for WT Texas H5 HA with a natural sialo-pentasaccharide from human milk LSTa (Neu5Ac α 2-3Gal β 1-3GlcNAc β 1-3Gal β 1-4Glc, referred to below as Sia-1, Gal-2, GlcNAc-3, Gal-4, and Glc-5) (Fig. 3A and table S1), which is an avian-type receptor analog (56). In the Texas H5 HA complex with LSTa, all five monosaccharide moieties are well ordered as indicated by clear, interpretable electron density (fig. S5, left), which may be impacted to some extent by a fortuitous crystal packing contact of the last monosaccharide moiety Glc-5 with a symmetry-related HA molecule (fig. S6). Structural comparisons were carried out for the LSTa complex of Texas H5 HA with H5N1 HA from A/Vietnam/1194/2004 [VN1194, clade 1, Protein Data Bank (PDB) ID 3ZP0] (57), which shares the same HA head domain with H5N1 HA from A/Vietnam/1203/2004 (Viet04, clade 1) (57) and H5N1 HA from A/Indonesia/5/2005 (Indo05, clade 2.1.3.2, PDB ID 4K63) (43) (Fig. 3, B to E). The Sia-1 (sialic acid) and Gal-2 (galactose) of LSTa bind to these three WT H5 HAs in a similar way, which contribute to specificity with avian-type receptors, where LSTa is in a trans conformation about the glycosidic bond between Gal-2 and Sia-1. The key avian H5 residue Gln²²⁶ makes hydrogen bonds with these two glycans (Fig. 3) (42). In VN1194 and Indo05 H5 HAs, the hydroxyl sidechain of Ser¹³⁷ hydrogen bonds to the Sia-1 carboxyl group, whereas in Texas H5 HA, the Ala¹³⁷ sidechain cannot form a hydrogen bond with LSTa (Fig. 3, A to C). The backbone carbonyl of Val¹³⁵, the backbone amide of residue 137, and the Glu¹⁹⁰ carboxyl also hydrogen bond with Sia-1 in all three structures. The last three carbohydrate moieties, GlcNAc-3 (N-acetylglucosamine), Gal-4, and Glc-5 (glucose) of LSTa, exit the binding pocket and make weak or no direct interactions with Texas H5 HA (Fig. 3A); GlcNAc-3 was found to be shifted from the corresponding GlcNAc-3 in Indo05 H5 HA complex (Fig. 3E).

Crystal structure of Texas H5 HA Gln²²⁶Leu with human receptor analog LSTc

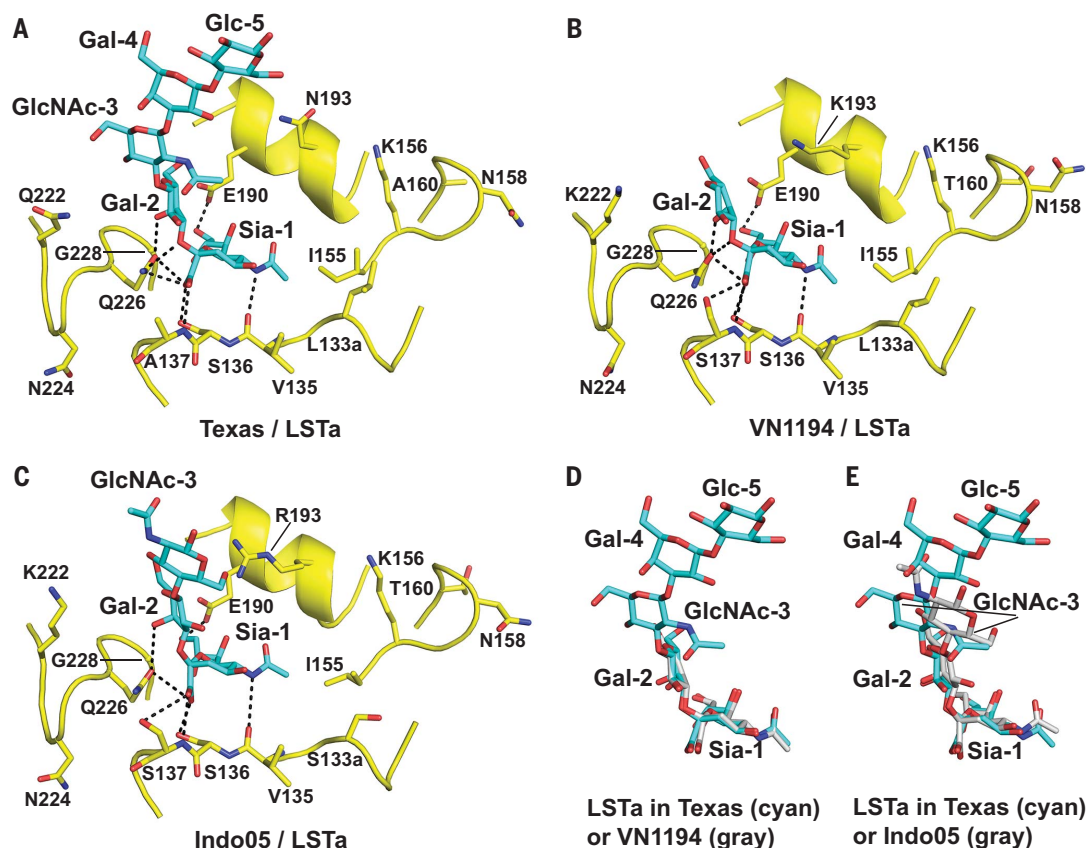
The Texas H5 HA Gln²²⁶Leu mutation alone was sufficient to switch the specificity from avian- to human-type receptors (Fig. 2 and figs. S1 and S2). We therefore determined the crystal structure at 2.7-Å resolution for Texas H5 HA Leu²²⁶ mutant with human-type receptor analog LSTc (Neu5Ac α 2-6Gal β 1-4GlcNAc β 1-

3Gal β 1-4Glc), which is also a natural sialo-pentasaccharide from human milk (Fig. 4A and table S1). Similar to the LSTa complex with Texas H5 HA, all five monosaccharide moieties of LSTc could be modeled (fig. S5, right). By comparison, in the LSTc complex with ferret-transmissible mutants of Viet04 HA (PDB ID 4KDO) (49) (Fig. 4B) and Indo05 HA (PDB ID 4K67) (43) (Fig. 4C), only the first three monosaccharides could be modeled. However, the first four monosaccharides were modeled in the LSTc complex with a Gln²²⁶Leu mutation of H5N1 HA (E5.1) from A/duck/Egypt/10185SS/2010 (dkEgy10, clade 2.2.1), which showed an avian- to human-type receptor specificity switch (58, 59) (Fig. 4D), likewise LSTc in complex with human H2N2 HA from A/Singapore/1/1957 (Fig. 4E) (57H2, PDB ID 2WR7) (60).

Consistent with the α -2-6 specificity and the crystal structures of these H5 mutants and H2 HAs, LSTc in the Leu²²⁶ mutant also adopts a cis conformation and binds Texas HA in a manner similar to dkEgy10 Q226L E5.1 HA and 57H2 HA but distinct from that of Viet04 and Indo05 mutants, especially in the trajectory and orientation of GlcNAc-3 (Fig. 4). In all these HA-LSTc complexes, the conformation of Sia-1 and Gal-2 is similar, with Leu²²⁶ making van der Waals' contacts with the nonpolar portions of LSTc; however, Ala¹³⁷ of bovine HA cannot make a sidechain hydrogen bond with Sia-1, unlike Ser¹³⁷ in the other three H5 structures, although its main chain amide makes a hydrogen bond in all structures (Fig. 4). In the Leu²²⁶ mutant, Lys¹⁵⁶ and Asn¹⁹³ make additional hydrogen bonds to the Glc-5 moiety of LSTc, which contribute to binding of LSTc to the Texas H5 HA (Fig. 4A). Although Lys¹⁵⁶ is conserved in the other three H5 HAs, Asn¹⁹³ of Texas HA is not conserved, Lys¹⁹³ is found in Viet04, and Arg¹⁹³ is found in both Indo05 and dkEgy10 (Fig. 4). In another study, a Lys¹⁹³Thr mutation in ferret-transmissible Viet04 and Indo05 mutants enhanced human-type receptor binding (53). It is noteworthy that Lys²²² makes a hydrogen bond with Gal-2 in dkEgy10 E5.1 HA and 57H2 HA, although it is too distant to make such a hydrogen bond with Gal-2 in Viet04 and Indo05 HA mutants. In Texas HA, Gln²²² makes no hydrogen bond to LSTc (Fig. 4).

In WT Viet04 and Indo05 H5 HAs, there is N-glycosylation of Asn¹⁵⁸, which is believed to reduce human-type receptor binding and was removed in the ferret-transmissible mutants

Fig. 3. Crystal structure of bovine Texas H5 HA with avian receptor analog LSTa and comparison with other H5 HA structures bound to LSTa. (A) RBS of WT Texas H5 HA (yellow carbon, blue nitrogen, and red oxygen atoms) bound with LSTa (cyan carbon, blue nitrogen, and red oxygen atoms). (B) RBS of VN1194 H5 HA bound with LSTa (PDB ID 3ZP0). (C) RBS of Indo05 H5 HA bound with LSTa (PDB ID 4K63). (D) Superposition of LSTa in Texas H5 HA and VN1194 H5 HA. (E) Superposition of LSTa in Texas H5 HA and Indo05 H5 HA. The HA structures are in a similar orientation after superimposition of their receptor binding subdomain (residues 117 to 265).



(49, 50). However, since around 2005, the 158 *N*-glycosylation of H5 HAs is rare (44), which is the case in dkEgy10 and Texas H5 HAs (Fig. 4). Combining this critical feature with the shorter side chain of Asn¹⁹³ (compared with Lys¹⁹³ or Arg¹⁹³), which hydrogen bonds with Glc-5, may enable the trajectory of LSTc to move closer to the 190 helix and 150 loop (Fig. 4A).

It is also noteworthy that the Texas and Viet04 mutant HAs have a Leu^{133a} insertion, and the Indo05 mutant HA a Ser^{133a} insertion, whereas no 133a insertions are present in dkEgy10 and 57H2 HAs (Fig. 4). The role of the 133a insertion in receptor binding has been investigated in previous studies with dkEgy10 H5 HA, which showed that insertion of Ser^{133a} abrogates receptor binding to both avian- and human-type receptors (58, 59), which is not the case for Texas H5 HA.

Conclusions

Whereas avian influenza H5N1 viruses impact the annual production of poultry and eggs, they also cause severe illness in marine and terrestrial mammals and pose a threat to human health owing to sporadic infections from domestic animals and the potential for the virus to emerge as a new pandemic strain in an immunologically naïve human popula-

tion (32). Since 2021, H5 subtype clade 2.3.4.4b viruses have become a dominant strain in wild birds, poultry, and dairy cows and have infected other mammalian species, including terrestrial and marine mammals as well as humans (13). However, to date, mutations in the RBS of clade 2.3.4.4b H5 HA that switch to human-type receptor specificity have not been reported in the literature. Through glycan binding studies and x-ray structure determination of recombinant bovine H5 HA, we have shown that the bovine H5N1 clade 2.3.4.4b virus retains specificity for avian-type receptors. Several other reports have similarly concluded that the WT bovine H5N1 exhibits only avian-type receptor specificity by analyzing either recombinant HA produced in mammalian cells or the whole virus by using assays identical or similar to those used in this study (51, 52, 61). A Thr¹⁹⁹Ile mutation in the HA of H5N1 viruses since 2023 was indeed reported to increase breadth to α 2-3-linked avian-type receptors for Texas H5 HA (62). Another report evaluating intact virus with an assay using polymeric sialoside ligands concluded that the bovine H5N1 exhibited avian-type specificity with weaker specificity for human-type receptors (63). By using the same assay, the distantly related Viet04 H5N1

virus as a reference showed dominant binding to avian-type receptors and weak binding to human-type receptors (63). All reports agree that the bovine H5N1 virus retains predominantly avian-type receptor specificity.

A switch from avian- to human-type receptor specificity is considered a major risk factor for transmission in humans (33, 35, 64). For this reason, our observation that the single Texas H5 HA Gln²²⁶Leu mutation can switch receptor specificity is a clear concern. The bovine airway and mammary gland epithelium have predominantly avian-type and, to a lesser extent, human-type receptors (65, 66), which provide little selective pressure for acquiring human-type receptor specificity. However, H5N1 infection of dairy workers, who have human-type receptors (34), provide selective pressure for human-type receptor specificity. Furthermore, in the flu season, coinfection with seasonal influenza viruses could lead to reassortment of bovine and human viruses and the creation of a hybrid virus that is more adapted for human infection.

Although the binding affinity of the Gln²²⁶Leu H5 HA mutant to the human-type receptors is relatively weak, the HAs of some human influenza viruses that have been able to transmit in the human population also exhibit weak

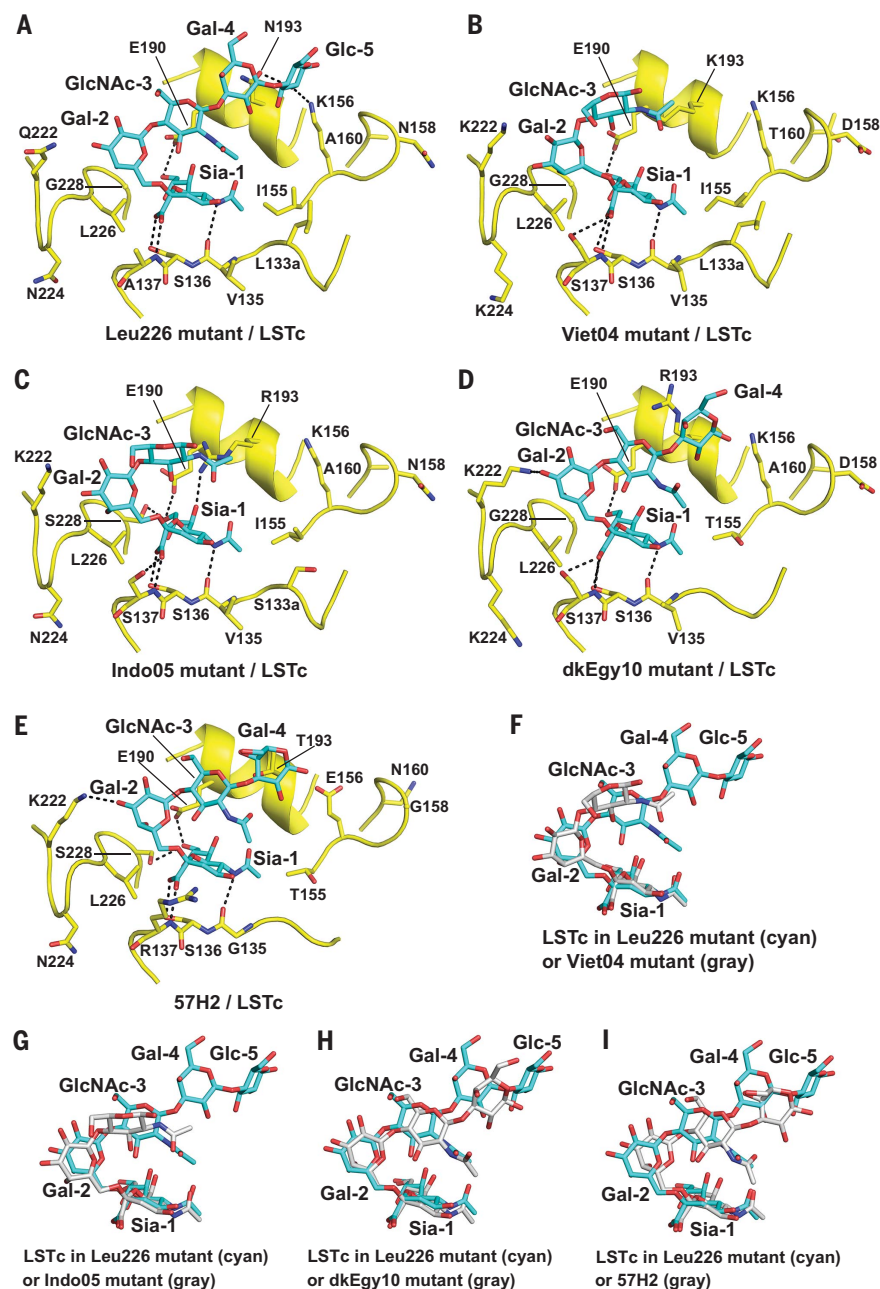


Fig. 4. Crystal structure of bovine Texas H5 HA Leu²²⁶ mutant with human receptor analog LSTc and comparison with other HA structures bound to LSTc. (A) RBS of Texas H5 HA Leu²²⁶ mutant (yellow carbon, blue nitrogen, and red oxygen atoms) bound to LSTc (cyan carbon, blue nitrogen, and red oxygen atoms). (B) RBS of Viet04 H5 HA ferret-transmissible mutant (Asn¹⁵⁸Asp/Asn²²⁴Lys/Gln²²⁶Leu/Thr³¹⁸Ile, H3 numbering) with LSTc (PDB ID 4KD0). (C) RBS of Indo05 H5 HA airborne ferret-transmissible mutant (His¹¹⁰Tyr/Thr¹⁶⁰Ala/Gln²²⁶Leu/Gly²²⁸Ser, H3 numbering) with LSTc (PDB ID 4K67). (D) RBS of dkEgy10 H5 HA human receptor preference E5.1 mutant (Gln²²⁶Leu) with LSTc (PDB ID 5E30). (E) RBS of 57H2 HA with LSTc (PDB ID 2WR7). (F) Superposition of LSTc in Texas HA Leu²²⁶ mutant and in a Viet04 H5 HA ferret-transmissible mutant. (G) Superposition of LSTc in Texas HA Leu²²⁶ mutant and in the Indo05 H5 HA ferret-transmissible mutant. (H) Superposition of LSTc in Texas HA Leu²²⁶ mutant and dkEgy10 H5 HA human receptor preferring mutant. (I) Superposition of LSTc in Texas HA Leu²²⁶ mutant and 57H2 HA. The HA structures are in a similar orientation after superimposition of their receptor binding subdomain (residues 117 to 265).

binding to human-type receptors (67). In particular, the HA from the 2009 “swine flu” H1N1 pandemic virus CA04 exhibited even lower avidity to biantennary N-linked glycan receptors in the ELISA-based assay than the Leu²²⁶-mutant H5 HA (Fig. 2B). The SPR data correspondingly showed weak binding of CA04 H1 HA for human-type receptors (Fig. 2A and fig. S2). With whole viruses, the relatively weak affinity observed with soluble HA trimers is amplified by the high density of HA molecules on the viral surface, affording increased avidity through simultaneous interactions with multiple receptors on the surface of the cell (68). Furthermore, with one additional mutation, Asn²²⁴Lys, or two additional mutations, Asn²²⁴Lys and Gly²²⁸Ser, the bovine H5 HA mutants Asn²²⁴Lys/Gln²²⁶Leu and Asn²²⁴Lys/Gln²²⁶Leu/Gly²²⁸Ser can bind more strongly to human-type receptors to achieve a level that is only slightly weaker than WT HA binding to avian-type receptors. Further studies are warranted for the phenotypic effects of such mutations to the H5N1 virus, such as virus entry, replication, and stability, among others (47). In addition to HA receptor specificity, other factors can affect transmission, such as a Glu⁶²⁷Lys mutation in polymerase basic 2 (PB2) protein (50, 69), and other human host factors, such as myxovirus resistance protein A (MxA), butyrophilin subfamily 3 member A3 (BTN3A3), and acidic nuclear phosphoprotein 32 (ANP32) (70), which are associated with viral adaptation in mammalian hosts and viral replication in human cells. Notably, Glu⁶²⁷Lys in PB2 has been detected in the first human-infecting Texas virus (26); otherwise, the sequence of Texas HA is highly similar to that of other bovine H5 HAs from cows and infected humans, with a sequence identity of 99.5 to 100% (figs. S7 and S8).

Our structural analyses and binding assays provide insights into the bovine H5 HA receptor binding and mutations that can switch specificity from avian to human receptors. Notably, a single Gln²²⁶Leu mutation can completely switch bovine H5 HA to human-type receptor specificity. In a recent deep mutational scanning study of a related avian H5N1 virus from A/American Wigeon/South Carolina/USDA-000345-001/2021, mutations at residue 226, including Leu²²⁶, were observed to increase binding to engineered α 2-6 versus α 2-3 293 cells (48). However, previous studies with most of the earlier H5 strains found that three or more mutations in the HA were usually required to switch receptor preference to from avian to human type (45, 49, 50, 54). Because each mutation is independent and the probability of achieving additional mutations decreases exponentially, our observation that a single mutation is sufficient to switch receptor specificity in the Texas HA dramatically increases the likelihood of achieving this

phenotype required for human transmission. In addition, mutations that could potentially affect receptor binding specificity have been found at RBS positions, including 135, 160, and 222, but they only occurred once, 13 times, and once, respectively, in 900 recent bovine H5 HA sequences (fig. S7). Thus, it is important to continue to monitor for signs of such changes in the currently circulating H5N1 viruses.

Limitations

In this study, we have not analyzed all the factors associated with human transmission of zoonotic influenza viruses. As discussed above, receptor specificity is a key factor, but not the only one required for human-to-human transmission of influenza viruses. The HA fusion pH is also important for transmission and HA stability (49). The pH of fusion of a recent HA from a bovine virus (A/dairy cattle/Texas/24-008749-001-original/2024) was determined to be 5.9, which is higher than that associated with efficient airborne transmission (pH < 5.5) (49, 71). The neuraminidase NA also plays a key role in infection, as well as the HA/NA balance related to receptor binding by the HA and receptor destroying activity of the NA. (72, 73).

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SUPPLEMENTARY MATERIALS

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Materials and Methods

Figs. S1 to S8

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MDAR Reproducibility Checklist

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Correction (5 December 2024): In the print abstract, there is an error in two places: "Glu226Leu" should be "Gln226Leu." This was corrected in the PDF and XML but was not caught soon enough to correct in print. This correction is to alert readers to the print error.



A single mutation in bovine influenza H5N1 hemagglutinin switches specificity to human receptors

Ting-Hui Lin, Xueyong Zhu, Shengyang Wang, Ding Zhang, Ryan McBride, Wenli Yu, Simeon Babarinde, James C. Paulson, and Ian A. Wilson

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Editor's summary

In 2021, a highly pathogenic influenza H5N1 clade 2.3.4.4b virus was detected in North America that is capable of infecting a diversity of avian species, marine mammals, and humans. In 2024, clade 2.3.4.4b virus spread widely in dairy cattle in the US, causing a few mild human cases, but retaining specificity for avian receptors. Historically, this virus has caused up to 30% fatality in humans, so Lin *et al.* performed a genetic and structural analysis of the mutations necessary to fully switch host receptor recognition. A single glutamine to leucine mutation at residue 226 of the virus hemagglutinin was sufficient to enact the change from avian to human specificity. In nature, the occurrence of this single mutation could be an indicator of human pandemic risk. —Caroline Ash

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